

TEST 5

Part 1

Question	Answer	Location	Explain
1	South West	Speaker B: I'm in the South West. Speaker A: Okay, and are you interested in having a regular cleaner or is this a one-off clean?	- Đáp án là South West - "I'm in the South West" → vị trí khách hàng - Khớp với "in the ..." trong đề - Key word: + South West (n): khu Tây Nam + area (n): khu vực
2	double	Speaker B: Well, one of the bedrooms is quite small, so it's probably really two double rooms. Speaker B: We have the small bedroom set up as an office.	- Đáp án là double - "two double rooms" → 2 phòng ngủ đôi - Khớp với đề - Key word: + double (adj): đôi + bedroom (n): phòng ngủ
3	office	Speaker B: We have the small bedroom set up as an office. Speaker A: Okay.	- Đáp án là office - "set up as an office" → dùng làm văn phòng - Key word: + office (n): văn phòng + set up (v): thiết lập
4	lounge	Speaker B: Then there's the lounge and as well as that we have a family room. Oh, and two bathrooms. Speaker A: Okay.	- Đáp án là lounge - Danh sách phòng: kitchen/diner + lounge + family room - Khớp chỗ trống - Key word: + lounge (n): phòng khách + family room (n): phòng sinh hoạt
5	A	Speaker A: We can put clean sheets on the beds, for example. Or we can clean out your fridge. A lot of people like that done. Speaker B: I'm not so worried about that, but I would like the sheets changed.	- Đáp án là A - Khách chọn "sheets changed" → thay ga giường - B sai vì không làm windows - C sai vì khách không chọn fridge - Key word: + sheets (n): ga giường + changed (v): thay

6	B	Speaker A: You can have it done every month if you like, but three months is what we usually recommend. Speaker B: That sounds good.	- Đáp án là B - “it” = carpet cleaning (shampoo carpet) - Làm kỹ hơn mỗi 3 tháng - Key word: + carpet (n): thảm + shampoo (v): giặt
7	B	Speaker B: I would like to have the ironing done. Speaker A: That's fine. And would you like the washing put away in drawers or wardrobes?	- Đáp án là B - “ironing done” → ủi quần áo - A sai vì khách tự giặt - C sai vì không cần cất đồ - Key word: + ironing (n): ủi đồ + washing (n): giặt
8	Amies	Speaker B: Yes, it's 48 Amies Road. That's A-M-Y-E-S. And that's in Mount Eden. Speaker A: OK.	- Đáp án là Amies - Đánh vần rõ ràng → A-M-Y-E-S - Key word: + address (n): địa chỉ + Road (n): đường
9	Thursday	Speaker A: We have a cleaner in your area later on Thursday, so shall we go for that day? Speaker A: We could have someone there by about 9.30 in the morning.	- Đáp án là Thursday - Chọn ngày thứ Năm - Key word: + Thursday (n): thứ Năm + cleaner (n): người dọn dẹp
10	3 hours	Speaker A: Well, you said the house isn't large, so I'd guess no more than three hours at the most. We can occasionally finish a three-bedroom in two and a half, but that's probably not too realistic.	- Đáp án là 3 hours - “no more than three hours” → tối đa 3 giờ - Key word: + hours (n): giờ + maximum (n): tối đa

You will hear a telephone conversation between a customer and a representative of an agency that arranges cleaning services for people's houses. First, you have some time to look at questions 1 to 4.

Now listen carefully and answer questions 1 to 4.

Speaker A: Good morning, Home Care Cleaning Agency.

Speaker B: Oh, hello. I'd like to arrange to have some cleaning done, please.

Speaker A: Certainly. Have you used our services before?

Speaker B: No, I haven't, but a friend of mine has often used your company and suggested I phone you.

Speaker A: Oh, that's great. I'll just take some details from you and then you can let me know exactly what you would like done.

Speaker B: Okay.

Speaker A: First of all, what area are you in?

Speaker B: I'm in the South West.

Speaker A: Okay, and are you interested in having a regular cleaner or is this a one-off clean?

Speaker B: No, I'd like to arrange to have someone come in once a week.

Speaker A: Right. Now, let's just get a bit of information about the house. How many bedrooms does it have?

Speaker B: Three.

Speaker A: Are they all double bedrooms?

Speaker B: Well, one of the bedrooms is quite small, so it's probably really two double rooms.

Speaker B: We have the small bedroom set up as an office.

Speaker A: Okay. And just a standard kitchen, lounge, dining room and bathroom?

Speaker B: Well, actually, we don't have a separate dining room. It's a kitchen-diner. Then there's the lounge and as well as that we have a family room. Oh, and two bathrooms.

Speaker A: Okay.

Before you hear the rest of the conversation, you have some time to look at questions five to ten.

Now listen and answer questions five to ten.

Speaker A: Right. That's all I need to know about the house itself. Now, what we do is we offer a basic cleaning service and then you let us know about anything extra you'd like cleaned.

Speaker B: Right. And what does the basic service include?

Speaker A: We clean all the floors and surfaces and do a basic clean in the kitchen and bathroom.

Speaker B: Okay, that sounds good. And do you clean windows?

Speaker A: Well, I'm afraid we don't do that. Health and safety regulations. But we do offer other special services. We can put clean sheets on the beds, for example. Or we can clean out your fridge. A lot of people like that done.

Speaker B: I'm not so worried about that, but I would like the sheets changed.

Speaker A: That's no problem. We can do that.

Speaker B: What about the carpet?

Speaker A: As well as the standard vacuum, we can shampoo it if you want.

Speaker B: Right. We've got young children, so it gets pretty dirty. I haven't had it done for six months. It really needs doing more often than that.

Speaker A: You can have it done every month if you like, but three months is what we usually recommend.

Speaker B: That sounds good. The only other thing is the washing. Do you do laundry work?

Speaker A: Yes, we do. That's no problem. What exactly would you like done?

Speaker B: Well, it's no problem to put a load of washing in the machine. I can do that, but I would like to have the ironing done.

Speaker A: That's fine. And would you like the washing put away in drawers or wardrobes?

Speaker B: No, it's OK. You can just leave everything out and fold it up.

Speaker A: OK, right. Now, I'll just set up an account for you and I'll take down some of your details. First of all, what's your full name?

Speaker B: It's Rose Killey.

Speaker A: And could I have your home address, please, Mrs Kelly?

Speaker B: Yes, it's 48 Amies Road. That's A-M-Y-E-S. And that's in Mount Eden.

Speaker A: OK. Now, what day of the week would you like the property cleaned?

Speaker B: It doesn't really matter. Not Monday, but any other day.

Speaker A: OK, let's see. We have a cleaner in your area later on Thursday, so shall we go for that day?

Speaker A: We could have someone there by about 9.30 in the morning. Does that suit you?

Speaker B: Yes, that's fine. And do you charge an hourly rate or a total for the whole job?

Speaker A: No, we charge by the hour. It will work out to about \$25 per hour.

Speaker B: Oh, that's quite reasonable. I thought it might be closer to \$30.

Speaker A: And how long do you expect it to take?

Speaker A: Well, you said the house isn't large, so I'd guess no more than three hours at the most. We can occasionally finish a three-bedroom in two and a half, but that's probably not too realistic.

Speaker B: Right. That sounds good.

Speaker A: OK, that's lovely. Thank you very much. And we'll have someone there next week.

Speaker B: Thank you.

That is the end of Part 1. You now have 45 seconds to check your answers to Part 1.

Part 2

Question	Answer	Location	Explain
11	C	<i>We had a big assembly where we were given a talk by a special former student, the internationally renowned author Jane Roberts. It was fascinating.</i>	- Đáp án là C - “a talk by a special former student... internationally renowned author” → mời người nổi tiếng - A sai vì không có viết bài hát mới - B sai vì không mở tòa nhà mới - Key word: + renowned (adj): nổi tiếng + former student (n): cựu học sinh
12	A	<i>And starting from this year, she'll be organizing a two-week trip to Japan for all students who would like to take part. We're very excited about this new development.</i>	- Đáp án là A - “organizing a two-week trip to Japan” → thêm chuyến đi nước ngoài - B sai vì đã có giáo viên Nhật từ trước - C sai vì visit từ học sinh Nhật là hoạt động cũ - Key word: + trip (n): chuyến đi + organizing (v): tổ chức
13	C	<i>I encourage you to check it often to see what work teachers have set for your child to do at home each week. This will help you to keep up with the tasks your children need to complete.</i>	- Đáp án là C - “see what work teachers have set” → xem bài tập về nhà - A sai vì order textbooks qua office - B sai vì giao tiếp trực tiếp với giáo viên - Key word: + work (n): bài tập + tasks (n): nhiệm vụ
14	A	<i>They performed really well in the regional competition last month, just missing out on first place. This didn't dampen their spirits, though, and they went on to raise over \$200 to give to the local children's hospital through fundraising in the school.</i>	- Đáp án là A - “raise over \$200... fundraising” → gây quỹ từ thiện - B sai vì không thiết kế áo - C sai vì không giành giải nhất - Key word: + raise money (v): gây quỹ + fundraising (n): gây quỹ

15	B	<i>The library, for example, always needs help with maintaining the records of the books we have in stock. That allows our wonderful teacher-librarian to concentrate on other things.</i>	- Đáp án là B - “maintaining the records” → giữ hồ sơ sách - A sai vì sửa sách do librarian - C sai vì trông trẻ cũng do librarian - Key word: + records (n): hồ sơ + maintain (v): duy trì
16	B	<i>We're very proud of the Performing Arts Centre, which you can see at the top of the map in the centre. It's just past the tennis courts.</i>	- Đáp án là B - “top... centre... past tennis courts” → vị trí B - Key word: + centre (n): trung tâm + tennis courts (n): sân tennis
17	D	<i>There are two drop-off bays along Fern Street, but please only use the first bay near the classrooms rather than the second one near the Sports Hall. That's for buses only.</i>	- Đáp án là D - “first bay near the classrooms” → vị trí gần classroom (D) - Key word: + bay (n): khu đỗ xe + classrooms (n): lớp học
18	C	<i>we now assemble on the netball courts, which is the oval shape you can see there in the top right-hand corner of the map. I'll point out the school canteen in case you want to join us for our family pancake breakfasts.</i>	- Đáp án là C - “oval shape... top right-hand corner” → điểm C - Key word: + oval (adj): hình bầu dục + assemble (v): tập trung
19	A	<i>It's an L-shaped building and can be easily accessed directly from the Sports Hall. For those of you needing IT help, such as for problems with school laptops, go to the IT Centre.</i>	- Đáp án là A - “L-shaped building... from Sports Hall” → vị trí A - Key word: + L-shaped (adj): hình chữ L + accessed (v): tiếp cận
20	H	<i>go to the IT Centre, which is to the left of the administration building. There's a covered walkway connecting it to the art block.</i>	- Đáp án là H - “left of administration building... near art block” → vị trí H - Key word: + left (n): bên trái + walkway (n): lối đi

Part 2. You will hear a talk given by a high school principal to parents whose children will soon be starting at the school. First you have some time to look at questions 11 to 15. Now listen carefully and answer questions 11 to 15.

Hello everyone and a very warm welcome to all our new parents and students to our information evening. Millbrook High School has a long history. In fact, we recently celebrated our 80th anniversary in this beautiful building, which dates back to the very early days of the school.

We had a big assembly where we were given a talk by a special former student, the internationally renowned author Jane Roberts. It was fascinating. She joined us in singing the school song and said it brought back very fond memories of singing those same words when she was a student here.

We are very proud of our academic achievements, but we're always striving to improve. There's an exciting change in our Japanese program that I want to tell you about this evening.

Since the program started five years ago, we've been fortunate enough to have a wonderful teacher from Japan. And starting from this year, she'll be organizing a two-week trip to Japan for all students who would like to take part. We're very excited about this new development, which is a nice addition to our traditional annual visit by school children from our sister school in Japan.

The school website is constantly being updated with the latest news. I encourage you to check it often to see what work teachers have set for your child to do at home each week. This will help you to keep up with the tasks your children need to complete.

Our teachers are more than happy to talk to parents, and you can organize a time to do that after class via the school office. The school office is also the place to take your textbook orders, by the way, and remember that they need to be in well before the start of school.

In terms of sport, we're very proud of how well our football team has done recently. They performed really well in the regional competition last month, just missing out on first place. This didn't dampen their spirits, though, and they went on to raise over \$200 to give to the local children's hospital through fundraising in the school.

We hope your child will be interested in joining one of our sporting clubs and proudly wearing the school team shirt.

We're always grateful if parents can help out at the school. The library, for example, always needs help with maintaining the records of the books we have in stock. That allows our wonderful teacher-librarian to concentrate on other things, such as making sure that any torn or damaged

books are fixed straight away, and looking after the children when they come in to borrow books. Let reception know if you're interested.

Before you hear the rest of the talk, you have some time to look at questions 16 to 20. Now listen and answer questions 16 to 20.

Now have a look at the map of the school, and I'll point out some places you might like to look around.

You came in via the administration building on Main Road, at the bottom of the map, and we're now in the Assembly Hall, just behind that.

We're very proud of the Performing Arts Centre, which you can see at the top of the map in the centre. It's just past the tennis courts. We put on all our school plays and musical events there.

Now a word about dropping off your children on school days. Always a busy time. There are two drop-off bays along Fern Street, but please only use the first bay near the classrooms rather than the second one near the Sports Hall. That's for buses only.

Another important area to point out is the fire evacuation point. We used to meet just behind the Assembly Hall, but as we've grown so big, we now assemble on the netball courts, which is the oval shape you can see there in the top right-hand corner of the map.

I'll point out the school canteen in case you want to join us for our family pancake breakfasts. It's an L-shaped building and can be easily accessed directly from the Sports Hall.

For those of you needing IT help, such as for problems with school laptops, go to the IT Centre, which is to the left of the administration building. There's a covered walkway connecting it to the art block.

We really hope your children...

That is the end of part two. You now have 30 seconds to check your answers to part two.

Part 3

Question	Answer	Location	Explain
21	B	Speaker B: I thought face-to-face interviews would be more appropriate because you can look at things in more depth when you're actually with the person. Speaker A: How many participants did you have?	- Đáp án là B - “look at things in more depth” → khám phá chi tiết - A sai vì không nói dễ hơn - C sai vì không liên quan đến khác biệt với nhóm khác - Key word: + depth (n): chiều sâu + appropriate (adj): phù hợp
22	C	Speaker B: I did was send round a really short questionnaire by email asking whether people liked maths and these 25 answered not at all, which is what I needed. Speaker B: Yeah, and arranged to interview them.	- Đáp án là C - Chọn người “answered not at all” → có cùng thái độ tiêu cực - A sai vì không phải số lượng - B sai vì không phải volunteer - Key word: + questionnaire (n): bảng khảo sát + attitude (n): thái độ
23	C	Speaker B: What did shock me was how violently anti-maths as a subject they were. It wasn't what I'd expected at all. Speaker A: But yes, I can imagine.	- Đáp án là C - “violently anti-maths” → cảm xúc rất mạnh - A sai vì không nói về self-assessment - B sai vì nói không có khác biệt giới tính - Key word: + shock (v): gây sốc + anti-maths (adj): ghét toán
24	teaching	Speaker B: Now you can see that for the women, they thought bad teaching was mainly to blame for their disliking maths. Speaker A: Didn't the men think that?	- Đáp án là teaching - “bad teaching” → nguyên nhân chính - Khớp “Poor ...” - Key word: + teaching (n): việc dạy học + blame (v): đổ lỗi
25	numbers	Speaker B: No, they talked mostly about the fact that they were actually, you know, scared of numbers. Speaker A: And some of these people considered themselves to be good at maths.	- Đáp án là numbers - “scared of numbers” → sợ con số - Khớp với “Fear of ...” - Key word: + numbers (n): con số + scared (adj): sợ

26	applications	Speaker B: the problem for the women at the second level was the fact that they couldn't see what applications maths had, what obvious ones anyway. Speaker A: Yes, that's been my experience as well.	- Đáp án là applications - “couldn't see what applications maths had” → không thấy ứng dụng - Key word: + applications (n): ứng dụng + obvious (adj): rõ ràng
27	talent	Speaker B: they had this strong belief that you have to be born with a talent for the subject to be any good at maths. Speaker A: I've come across that quite often in my work too.	- Đáp án là talent - “born with a talent” → cần năng khiếu - Key word: + talent (n): năng khiếu + belief (n): niềm tin
28	observation	Speaker B: Well, the method I used for my research was observation. Speaker A: Of what?	- Đáp án là observation - Nói trực tiếp phương pháp - Key word: + observation (n): quan sát + method (n): phương pháp
29	whole	Speaker B: they attempted to comprehend the question taken as a whole. This obviously helped them a great deal.	- Đáp án là whole - “taken as a whole” → hiểu tổng thể - Key word: + whole (n): tổng thể + comprehend (v): hiểu
30	reading	Speaker B: Well, I have to say the strategy they used was to give up immediately after reading the question, I'm afraid. Speaker A: Well, that's extremely interesting, Aya.	- Đáp án là reading - “after reading the question” → bỏ cuộc sau khi đọc - Key word: + reading (n): đọc + give up (v): bỏ cuộc

Part 3

You will hear a student called Aya discussing her research project about math's ability with her tutor. First you have some time to look at questions 21 to 23. Now listen carefully and answer questions 21 to 23.

Speaker A: Right, Aya, last time we met you decided to choose skills and attitudes in learning maths as the focus of your research project and you were thinking of splitting it into two sections, is that right?

Speaker B: Yes, I decided to start by looking at people's attitude to maths and then go on to research how people actually solve different maths problems.

Speaker A: Great. Now what research methods did you use?

Speaker B: Well, I explored lots of different techniques and actually tried out things like sending out questionnaires but then I opted for a different approach. I thought face-to-face interviews would be more appropriate because you can look at things in more depth when you're actually with the person.

Speaker A: How many participants did you have?

Speaker B: Quite a few, which meant interviewing was pretty demanding work. 25 altogether, 14 men and 11 women.

Speaker A: How did you identify them?

Speaker B: Well, they were all students at this college and what I did was send round a really short questionnaire by email asking whether people liked maths and these 25 answered not at all, which is what I needed.

Speaker A: So you followed them up?

Speaker B: Yeah, and arranged to interview them.

Speaker B: Now the first question I asked them was, how good do you think you are at maths? And I got these results. A pretty even split at either end of the continuum. 20% for both good and poor and the majority in the middle.

Speaker B: And I should emphasise that for this there were no distinct differences between men and women.

Speaker A: Did those figures surprise you?

Speaker B: I wouldn't say so. What did shock me was how violently anti-maths as a subject they were. It wasn't what I'd expected at all. But yes, I can imagine.

Before you hear the rest of the discussion, you have some time to look at questions 24 to 30. Now listen and answer questions 24 to 30.

Speaker B: Anyway, the next part of my interview asked about their experiences of learning maths and why they had become so negative about it. I identified three different levels of importance, as you can see in this table.

Speaker A: Do you mean the first row shows those points which were considered most serious?

Speaker B: Yes, that's right. And the second row, the second most important, and so on. And I also showed the difference between female and male responses.

Speaker B: Now you can see that for the women, they thought bad teaching was mainly to blame for their disliking maths.

Speaker A: Didn't the men think that?

Speaker B: No, they talked mostly about the fact that they were actually, you know, scared of numbers.

Speaker A: And some of these people considered themselves to be good at maths.

Speaker B: Yeah, interesting, isn't it?

Speaker B: And then the problem for the women at the second level was the fact that they couldn't see what applications maths had, what obvious ones anyway.

Speaker A: Yes, that's been my experience as well.

Speaker B: And for the men, they found problems because they had this strong belief that you have to be born with a talent for the subject to be any good at maths.

Speaker A: I've come across that quite often in my work too.

Speaker B: And then at the third level, it was striking that men and women had the same problem, which was that they weren't really aware of alternative strategies, you know, for solving problems.

Speaker A: Thanks.

Speaker A: Now, what about the second part of your study, Aya?

Speaker B: Well, the method I used for my research was observation.

Speaker A: Of what?

Speaker B: Of how different people solve maths problems. So basically, I set them a problem and asked them to write down any workings out and I made notes on how they did it.

Speaker A: Good mathematicians?

Speaker B: A mixture, actually. Five successful and five less successful.

Speaker B: Now, I found that the most common strategy used by the people who got the correct answers was that they attempted to comprehend the question taken as a whole. This obviously helped them a great deal, in contrast with the way unsuccessful participants worked.

Speaker B: For them, the most common strategy was to take a solution which they had learned before and try to apply it.

Speaker A: And were there any other strategies?

Speaker B: Yes. The second most common technique I saw used by the good problem solvers was visualisation, which...

Speaker A: How interesting. I've noticed that's actually how I work best.

Speaker B: Yep.

Speaker A: And what about the weaker ones?

Speaker B: Well, I have to say the strategy they used was to give up immediately after reading the question, I'm afraid.

Speaker A: Well, that's extremely interesting, Aya. Now, had you thought about...

That is the end of Part 3. You now have 45 seconds to check your answers to Part 3.

Part 4

Question	Answer	Location	Explain
31	plants	<i>It grazes on plants. It's an herbivore, in other words. Here in North America, we can find it all along our west coast, from Alaska down to California.</i>	- Đáp án là plants - “grazes on plants” → ăn thực vật - Khớp với “feeds on ...” - Key word: + graze (v): ăn cỏ + herbivore (n): động vật ăn cỏ
32	spines	<i>And it survives very well because it's covered in spines, which are very sharp, and these deter most predators. Now, in the 1960s, in this country, the Sea Urchin was thought to be partly responsible for the sudden decline in kelp.</i>	- Đáp án là spines - “covered in spines” → được bao phủ bởi gai - Key word: + spines (n): gai + predators (n): động vật săn mồi
33	pest	<i>So, in fact, it was considered to be a pest. Incidentally, since the 1960s, the thinking on that has changed.</i>	- Đáp án là pest - “considered to be a pest” → bị coi là sinh vật gây hại - Key word: + pest (n): sinh vật gây hại + considered (v): được coi là
34	Asia	<i>It became apparent that there was a demand for the Red Sea Urchin in Asia, where certain parts of the animal were considered to be a great delicacy. And so, a commercial fishery was established based on Sea Urchins.</i>	- Đáp án là Asia - “demand... in Asia” → thị trường ở châu Á - Key word: + demand (n): nhu cầu + delicacy (n): món ngon

35	decline	<i>And just at this time, people noticed an apparent decline in the numbers of Sea Urchins. This prompted a wave of scientific studies.</i>	- Đáp án là decline - “decline in the numbers” → sự suy giảm - Khớp với đề - Key word: + decline (n): suy giảm + numbers (n): số lượng
36	200 years	<i>They can live to be 100 years old, and some may even reach 200. Just imagine, an individual Red Sea Urchin that hatched early in the 19th century may still be thriving.</i>	- Đáp án là 200 years - “reach 200” → tuổi tối đa 200 năm - Key word: + reach (v): đạt tới + years (n): năm
37	0.1 cm	<i>And from the age of 22, when it's about 12 centimeters long, it grows at just 0.1 of a centimeter a year. But somehow, the animal never really stops growing.</i>	- Đáp án là 0.1 cm - “0.1 of a centimeter a year” → tốc độ tăng trưởng - Key word: + centimeter (n): cm + grow (v): phát triển
38	aging	<i>the study measured levels of radiocarbon, and it has shown that even very old Sea Urchins show almost no sign of aging when something finally kills them. Unlike when someone looks at you or me.</i>	- Đáp án là aging - “no sign of aging” → không có dấu hiệu lão hóa - Key word: + aging (n): lão hóa + sign (n): dấu hiệu
39	breed	<i>even when they're incredibly ancient, they are still able to breed. So by now, you may be wondering how.</i>	- Đáp án là breed - “still able to breed” → vẫn sinh sản - Key word: + breed (v): sinh sản + ancient (adj): rất già
40	ocean conditions	<i>The study also confirmed the finding that in older Sea Urchins, there is a very steady, very consistent rate of growth. And this growth is independent of ocean conditions and other variables.</i>	- Đáp án là ocean conditions - “independent of ocean conditions” → không phụ thuộc điều kiện môi trường - Key word: + conditions (n): điều kiện + independent (adj): độc lập

Part 4. You will hear part of a lecture about a marine creature called the Red Sea Urchin. First, you have some time to look at questions 31 to 40. Now, listen carefully and answer questions 31 to 40.

Good morning. Last week, I talked about various research methods in the natural sciences, and you did some background reading on that topic.

Today, I want to take up that theme again and look at a specific example of how data gained using newer techniques has really pushed our knowledge forward, more or less overturning existing theories.

So, I'm going to look at the case of the Red Sea Urchin.

First of all, some background. As the name suggests, this is a marine animal, and it's found in shallow coastal waters. It grazes on plants. It's an herbivore, in other words.

Here in North America, we can find it all along our west coast, from Alaska down to California, so many of you will be familiar with it. And it survives very well because it's covered in spines, which are very sharp, and these deter most predators.

Now, in the 1960s, in this country, the Sea Urchin was thought to be partly responsible for the sudden decline in kelp, a plant which supports an extensive ecosystem. So, in fact, it was considered to be a pest.

Incidentally, since the 1960s, the thinking on that has changed, and humans are now thought to be to blame. But anyway, Red Sea Urchins weren't popular, and people even tried to poison them to get rid of them in any way possible.

But then, in the 1970s, attitudes changed. Interestingly, this had nothing to do with science. There were no new insights into the biology of the Sea Urchin. It was, in fact, a purely economic development.

It became apparent that there was a demand for the Red Sea Urchin in Asia, where certain parts of the animal were considered to be a great delicacy. And so, a commercial fishery was established based on Sea Urchins. They brought high prices.

In fact, at one point during the 1990s, they were one of the most valuable marine resources in the state of California. So, people began to look at the Sea Urchin in a different light.

And just at this time, people noticed an apparent decline in the numbers of Sea Urchins. This prompted a wave of scientific studies with the purpose of getting more information about the species—its life cycle, its biology, survival rate, growth patterns, and so on.

So, what did research uncover? Well, the most significant findings came from research based on biochemical techniques. Basically, individual Sea Urchins were tagged and injected with tetracycline, which allowed researchers to track their growth rates.

And it quickly became apparent that the Sea Urchins grew a lot more slowly and lived a lot longer than had previously been thought. Astonishingly, they can have a vast lifespan, greater than that of almost any other terrestrial or marine species. They can live to be 100 years old, and some may even reach 200.

Just imagine, an individual Red Sea Urchin that hatched early in the 19th century may still be thriving and even breeding.

It was found that this animal does grow fairly quickly when it's young. At the age of 2, it can grow from 2 to 4 centimeters in just one year. But even then, according to scientists who conducted the research, it still takes 6 or 7 years before it's big enough to be harvested.

By the time the Red Sea Urchin is a teenager, its growth slows quite dramatically. And from the age of 22, when it's about 12 centimeters long, it grows at just 0.1 of a centimeter a year.

But somehow, the animal never really stops growing. It just grows incredibly slowly. Some of the largest and probably the oldest Sea Urchins were found off the coast of British Columbia, between Vancouver Island and the mainland.

And now, another very recent study, one using an entirely different technique, has produced independent evidence for these findings. The study measured levels of radiocarbon, and it has shown that even very old Sea Urchins show almost no sign of aging when something finally kills them.

Unlike when someone looks at you or me, you can't look at a Sea Urchin and say whether it's young or middle-aged. And in addition to that, even when they're incredibly ancient, they are still able to breed.

So by now, you may be wondering how, with this phenomenal lifespan, there aren't a lot more Sea Urchins around than there actually are. Well, there is a suggestion that reproduction rates for the species are low, because not many Red Sea Urchins are able to survive the first year of life.

The study also confirmed the finding that in older Sea Urchins, there is a very steady, very consistent rate of growth. And this growth is independent of ocean conditions and other variables.

So, what are the implications of these findings?

That is the end of Part 4. You now have 1 minute to check your answers to Part 4. That is the end of the listening test. You now have 10 minutes to transfer your answers to the listening answer sheet. You have 2 minutes left. You have 1 minute left. Please stop writing and wait for your question booklet to be collected.

